POLITECNICO MILANO 1863

Master of Science Thesis in Management Engineering

When Does Model Complexity Pay Off?

A Comparison of Temporal and Aggregated EEG Representations for Advertising Recall Prediction

Supervisor: Lucio Lamberti

Co-supervisor: Marc-Antoine Fortin

Candidate: Amirhossein Jandaghian

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The Battle for Attention:
Massive Investment, Limited Attention

Attention Crisis

Consumers are exposed to **thousands of advertisements every day**, creating **advertising clutter** and **attention fatigue**, where many ads fail to remain in memory.

Massive Advertising Investment

Advertising is a **multi-billion-euro** industry, with **€9.6 billion** spent in **Italy** in **2025**, making campaign effectiveness a major financial concern for companies. (**40%** in linear TV)

The Strategic Challenge

Companies therefore need **more reliable methods to evaluate advertising effectiveness before launching campaigns**, in order to reduce the risk of costly ineffective ads.

The Shift Toward Neuromarketing

Traditional Methods

- Post-exposure
- Conscious self-reporting
- Inherently biased

Shift

Neuromarketing

- Real-time measurement
- Objective
- Physiological signals (e.g., EEG)

Key Questions for Advertisers

In the creative development phase:

1. Which specific factors and features make an ad memorable?

In the pre-launch testing phase:

2. Will this advertisement remain in memory or become a multi-million-euro mistake?

Research Questions

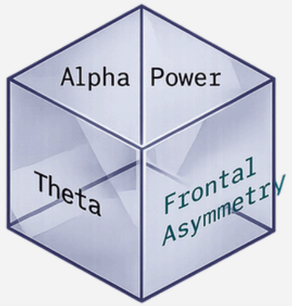
How do **EEG representations** affect the **prediction of advertising memorization?**

RQ1.
Prediction Performance
Do temporal representation methods outperform classic handcrafted EEG features?

RQ2.
Efficiency
What is the trade-off between computational cost, interpretability, and predictive gains?

EEG Representation Strategies

Handcrafted Features



- Aggregated statistics
- Simple representation
- Interpretable

Classic Mean

Deep Learning Embeddings

- Learns patterns inside time series
- Preserves neural dynamics
- Higher model complexity

TS2Vec

FEMBA

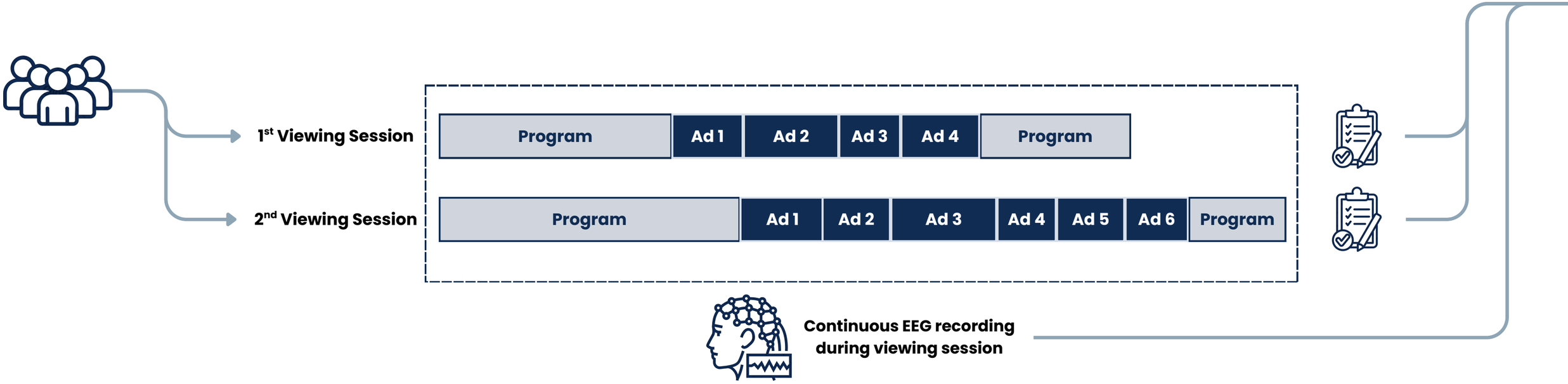
Experimental Setup

Experiment Location
BRIEL Lab
Politecnico di Milano

Participants
300 in Linear TV condition
Aged 18 - 80

Content
Editorial video content interrupted by ad breaks (for 13 brands)

Data Capture
Sync 8-channel EEG recording (10-20 placement)



Analytical Pipeline

A structured workflow enabling fair and reproducible comparison under identical modeling conditions

Step 01

Ensure quality and a harmonized dataset via filtering and validation

Data Cleaning

Step 02

Construct the modeling dataset and **generate EEG representations**

Feature Engineering

Step 03

Select the best-performing baseline model **[Gradient Boosting]**

Model Selection

Step 04

Compare predictive performance and computational efficiency.

Comparative Evaluation

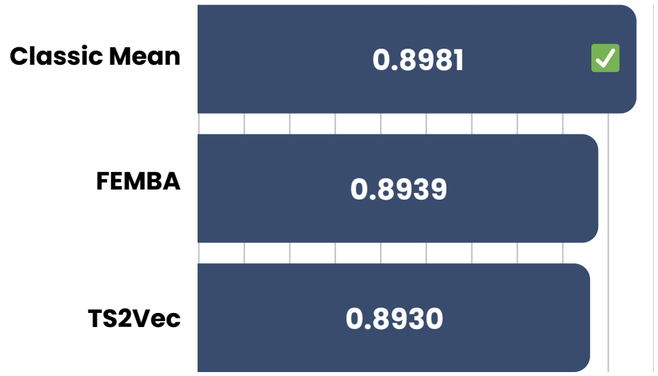
Comparative Results

All representations achieve strong predictive performance, achieving 83% accuracy and ~0.90 ROC-AUC.

Overall **brain engagement aggregated** over time captures the **dominant signal** driving **memorization** (Classic Mean), while **temporal models** capture **additional dynamics** (Embeddings).

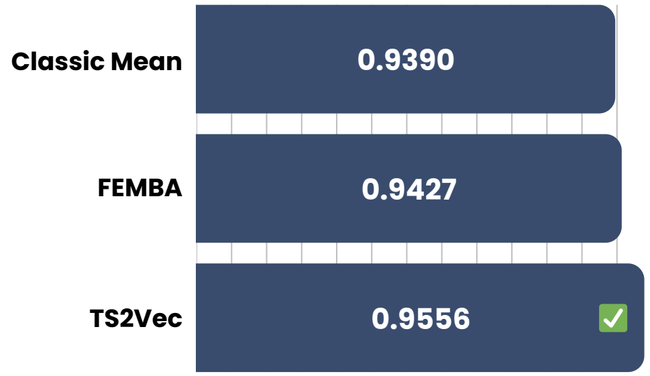
Temporal models require higher training time, but the **cost remains manageable;** Inference times are consistently low.

ROC - AUC



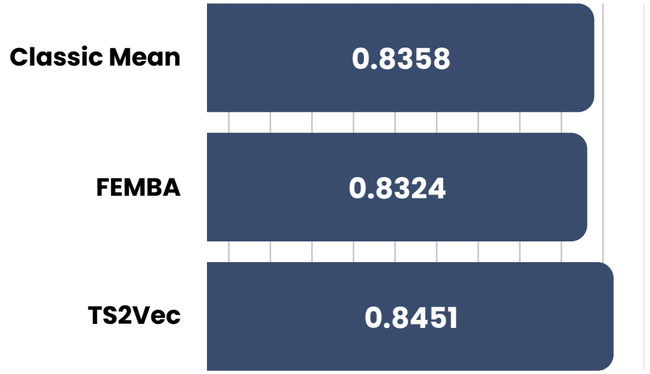
Nearly **identical** discrimination performance

Recall



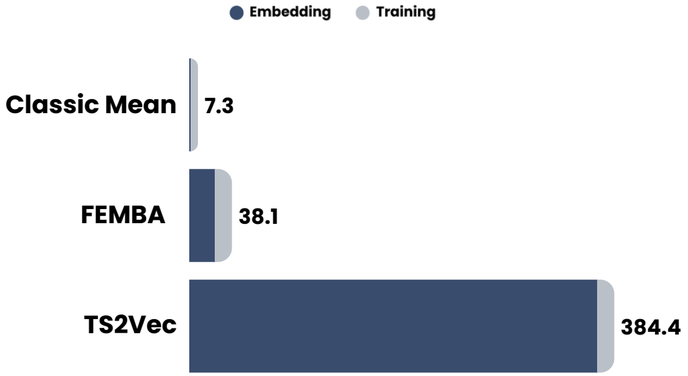
Higher sensitivity in detecting recalled ads byTS2Vec (**+1.6%**)

Accuracy



Consistently strong accuracy across representations

Computational Cost (Seconds)



Under 7 minutes total; Operationally negligible cost

Managerial Implication

EEG-based prediction enhances **advertising decision quality** and **reduces investment risk.**

Higher Recall enables better detection of high-performing ads, leading to more efficient allocation of advertising budgets and increased profitability:

Incremental Value ≈
Δ Probability of selecting high-performing ads **×**
Δ ROAS (Top vs Average Ads)

Model Complexity is a **Strategic Choice.**

Since both approaches perform well, higher complexity does not always mean better performance.
In our case, the complex model is justified because:

In large-scale campaigns, even marginal improvements in selection accuracy can translate into substantial financial impact.

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Thank You!

03.26.2026 | Amirhossein Jandaghian

Comprehensive Comparison of EEG Representations Across Performance and Computational Cost

Representation	Performance					Cost & Complexity		
	ROC-AUC	F1-score	Recall	Precision	Accuracy	Embedding Time (s)	Training Time (s)	Interpretability
Classic Mean	**0.8981**	0.8774	0.939	0.8233	0.8358	-	6.3	**High**
TS2Vec	0.893	**0.8853**	**0.9556**	**0.8246**	**0.8451**	369.9	15.4	Low
FEMBA	0.8939	0.8755	0.9427	0.8173	0.8324	23.8	15.3	Low

How We Estimated the Extra Profit:

Incremental Value ≈
Δ Probability of selecting high-performing ads

Any Method used for now (either focus groups, surveys, etc): P1
–
Prediction with Model (Recall = 95%): P2

×
Δ ROAS (Top vs Average Ads)

Assuming two good ads with two different returns as if they were published:
Return of Better Ads (R_high) > Return of Good Ad (R_avg)

Expected Value:

Baseline (traditional methods): $E[Return_baseline] = P1 \times R_high + (1 - P1) \times R_avg$
With model: $E[Return_model] = P2 \times R_high + (1 - P2) \times R_avg$

$$\begin{aligned} \text{Incremental Value} &= E[Return_model] - E[Return_baseline] \\ &= (P2 - P1) \times (R_high - R_avg) \\ &= \mathbf{\Delta P \times \Delta ROAS} \end{aligned}$$

This formulation assumes a simplified binary outcome (high-performing vs average ads).

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Contatti

+39 379 1456 213
amirhossein.jandaghian@mail.polimi.it
linkedin.com/in/amirh-jandaghian/